

QUESTION 1 Use a *separate* piece of paper

Marks

- a) Find the value of $\frac{6}{5 \times 7 - 4}$ correct to 3 significant figures. 2
- b) By rationalising the denominator, express $\frac{6}{\sqrt{7}+2}$ in the form $a\sqrt{7}-b$. 2
- c) Evaluate $\lim_{x \rightarrow -2} \frac{x^2 + x - 2}{x + 2}$ 2
- d) Differentiate the following with respect to x : 9
- (i) e^{7x}
- (ii) $\frac{5}{x^2}$
- (iii) $x^2 \log x$
- (iv) $\frac{e^x}{7x-3}$
- e) Find: 9
- (i) $\int (3x^2 - 2x + 1) dx$
- (ii) $\int \sqrt{6x+5} dx$
- (iii) $\int_0^1 e^{2x} dx$

QUESTION 2 Use a *separate* piece of paper

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A, B and C are the points (5,0), (8,4) and (0,10) respectively.

- (i) On a number plane draw $\triangle ABC$
- (ii) Find the slope of the line BC.
- (iii) Prove that $AB \perp BC$.
- (iv) Calculate the length of AB.
- (v) Prove $\triangle ACO \equiv \triangle ACB$, giving reasons for your answer. ✓
- (vi) Calculate the area of quadrilateral ABCO, where O is the origin.

QUESTION 3 Use a *separate* piece of paper

Marks

- a) The probability that Nathan will be present in class on any day is $\frac{3}{4}$ and the probability that Matthew is present is $\frac{19}{20}$. What is the probability that : 4
- (i) they will both be present on any particular day?
- (ii) at least one of them will be absent from class on any particular day?

- b) Use Simpson's Rule to find an approximation for $\int_1^3 f(x) dx$ using the values in the table below: 3

	1	2	3	4	5
x	1	1.5	2	2.5	3
f(x)	8.6	11.9	23.7	39.8	56.7

- c) A parabola has the equation $y + 11 = (x + 3)^2$. Find : 4
- (i) the coordinates of the vertex.
- (ii) the focal length.
- (iii) coordinates of the focus.

QUESTION 4 Use a *separate* piece of paper

- a) The 6th term of an arithmetic sequence is 58 and the 9th term is 112. Find : 7
- (i) the common difference.
- (ii) the first term.
- (iii) the sum of the first 100 terms of the sequence.

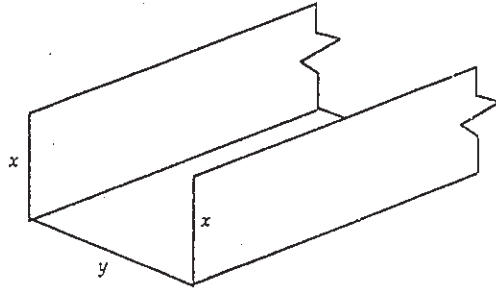
- b) Find the sum of the infinite geometric series $6 + 2 + \frac{2}{3} + \dots$ 3

- c) Find the equation of the tangent to the curve $y = e^{2x} + x$ at the point (0,1). 4

QUESTION 5 Use a *separate* piece of paper

Marks

- a) Find the volume of the solid formed when the curve $y = x^2 - 1$ is rotated about the y axis from $y = -1$ to $y = 3$. 4
- b) A home guttering company makes metal gutters from material which is 36 cm wide. The gutter is open at the top and it has a rectangular cross section. 8



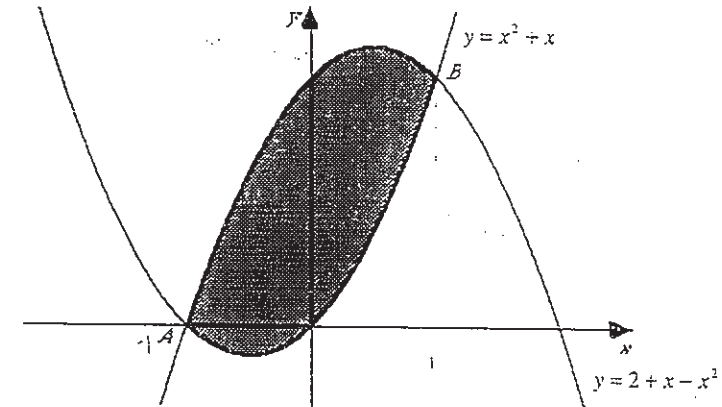
- (i) Show that $y = 36 - 2x$
- (ii) Show that the area, $A \text{ cm}^2$, of the rectangular cross section is given by $A = 36x - 2x^2$
- (iii) Find the value of x for which the area will be a maximum

QUESTION 6 Use a *separate* piece of paper

Marks

- a) Solve $(x+3)^2 + 5(x+3) + 6 = 0$ 4
- b) If α and β are roots of the equation $3x^2 - 2x - 1 = 0$, find the value of: 4
- (i) $\alpha + \beta$
- (ii) $\alpha\beta$
- (iii) $\alpha^2 + \beta^2$

c)



The diagram shows the curves $y = x^2 + x$ and $y = 2 + x - x^2$.

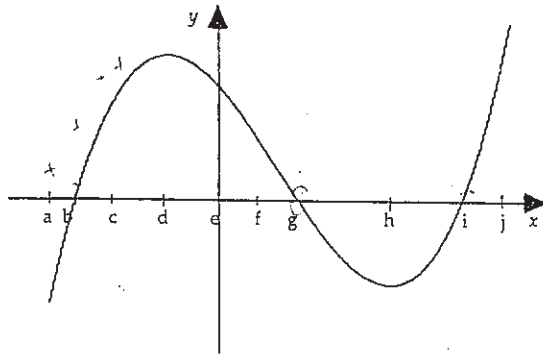
- (i) Find the coordinates of A and B.
- (ii) Calculate the area of the shaded region.

QUESTION 7 Use a *separate* piece of paper

Marks

a) The diagram shows the graph of $y = f(x)$

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(i) At what values of x on the function is :

(α) $f(x) = 0$?

(β) $f'(x) = 0$?

(γ) $f''(x) = 0$?

(ii) Between which values of x on the function is :

(α) the first derivative positive ?

(β) the second derivative negative ?

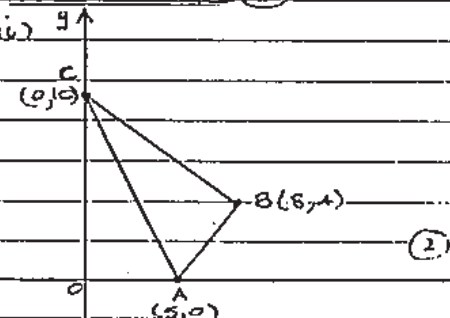
(γ) the function decreasing ?

(iii) Sketch the graph of $y = f'(x)$.

b) Given that $\frac{d}{dx} \{(\log x)^2\} = \frac{2 \log x}{x}$, evaluate $\int_1^e \frac{\log x}{x} dx$

3

a) $\frac{6}{5\sqrt{7}-4} = 0.194518387$
 $\frac{6}{5\sqrt{7}-4} = 0.194 \text{ (to 3 sig figs)}$
 b) $\frac{6}{\sqrt{7}+2} \times \frac{\sqrt{7}-2}{\sqrt{7}-2}$
 $= \frac{6(\sqrt{7}-2)}{7-4}$
 $= \frac{6\sqrt{7}-12}{3}$
 $= 2\sqrt{7}-4$ (2)
 c) $\lim_{x \rightarrow -2} \frac{x^2+x-2}{x+2}$
 $\lim_{x \rightarrow -2} \frac{(x+2)(x-1)}{(x+2)}$
 $= \lim_{x \rightarrow -2} (x-1)$
 $= -2-1$
 $= -3$ (2)
 d) (i) $f(x) = e^{-1/x}$
 $f'(x) = 7e^{-1/x}$ (1)
 (ii) $f(x) = \frac{5}{x^2}$
 $f'(x) = \frac{-5(2x)}{x^4}$
 $= -\frac{10}{x^3}$ (2)
 (iii) $f(x) = x^2 \log x$
 $f'(x) = (x^2)(\frac{1}{x}) + (\log x)(2x)$
 $= x + 2x \log x$ (3)
 (iv) $f(x) = \frac{7x-3}{e^x}$
 $f'(x) = \frac{(7x-3)(e^x) - (e^x)(7)}{(e^x)^2}$
 $= \frac{7xe^x - 3e^x - 7e^x}{(e^x)^2}$
 $= \frac{7xe^x - 10e^x}{(e^x)^2}$ (3)

(i) $\int \sqrt{6x+5} dx$
 $= \int (6x+5)^{\frac{1}{2}} dx$
 $= \frac{(6x+5)^{\frac{3}{2}}}{(\frac{3}{2})(6)} + c$
 $= \frac{(6x+5)^{\frac{3}{2}}}{9} + c$ (3)
 $= \frac{1}{9}(6x+5)\sqrt{6x+5} + c$
 (ii) $\int_0^1 e^{2x} dx$
 $= [\frac{1}{2}e^{2x}]_0^1$
 $= \frac{1}{2}e^2 - \frac{1}{2}e^0 = \frac{1}{2}e^2 - \frac{1}{2}$ (3)
 Question 2 (13)
 (i)  (2)
 (ii) $m_{BC} = \frac{4-0}{8-5}$
 $= \frac{4}{3}$ (2)
 (iii) $m_{AB} = \frac{0-9}{5-0}$
 $= -\frac{9}{5}$
 $m_{AB} \cdot m_{BC} = \frac{4}{3} \times -\frac{9}{5} = -1$
 $\therefore AB \perp BC$ (2)
 (iv) $d_{AB} = \sqrt{(8-5)^2 + (4-0)^2}$
 $= \sqrt{3^2 + 4^2}$
 $= \sqrt{25}$
 $= 5 \text{ units}$ (2)

AC = CA (common side) (1)
 OA = AB = 5 units (given) (s)
 $\therefore \triangle AOC \equiv \triangle AOB$ (RHS) (3)
 (vi) Area ABCO = 2 x Area $\triangle AOC$
 $= 2 \times \frac{1}{2} \times 5 \times 10$
 $= 50 \text{ units}^2$ (2)
 Question 3 (18)
 a) (i) $P(\text{both present}) = \frac{3}{4} \times \frac{19}{20}$
 $= \frac{57}{80}$ (2)
 (ii) $P(\text{at least 1 absent})$
 $= 1 - P(\text{both present})$
 $= 1 - \frac{57}{80}$
 $= \frac{23}{80}$ (2)
 b) $\int_0^8 f(x) dx = 3 \{ y_1 + 4y_2 + 2y_3 + y_4 \}$
 $= \frac{3}{3} \{ 8 \cdot 6 + 4(11 \cdot 9 + 39 \cdot 8) + 2(23 \cdot 7) + 56 \cdot 7 \}$
 $= 53.25$ (3)
 c) $y = e^{2x} + x$
 $\frac{dy}{dx} = 2e^{2x} + 1$
 when $x = 0$, $\frac{dy}{dx} = 2e^0 + 1 = 3$
 $\therefore$ slope of tangent is 3
 $y - 1 = 3(x - 0)$
 $y - 1 = 3x$
 $3x - y + 1 = 0$
 Question 4 (12)
 a) $V = \pi \int x^2 dy$
 $y = x^2 - 1$
 $x^2 = y + 1$
 $\therefore V = \pi \int_{-1}^3 (y+1) dy$
 $= \pi [\frac{1}{2}y^2 + y]_{-1}^3$
 $= \pi [\frac{1}{2}(3)^2 + 3 - \frac{1}{2}(-1)^2 - (-1)]$
 $= \pi (\frac{9}{2} + 3 - \frac{1}{2} + 1)$
 $= 8\pi \text{ units}^3$ (4)
 Question 5 (14)
 a) $T_b = 58$ $T_a = 112$
 $a + 5d = 58$ $a + 9d = 112$
 $a + 5d = 58$
 $a + 9d = 112$
 $3d = 54$
 $d = 18$
 $a = 58 - 5d = 58 - 90 = -32$
 $\therefore$ common difference is 18 (3)

$a + 5(18) = 58$
 $a + 90 = 58$
 $a = -32$
 $\therefore$ first term is -32
 (ii) $S_n = \frac{n}{2} [2a + (n-1)d]$
 $S_{100} = \frac{100}{2} [2(-32) + 99(18)]$
 $= 50(1718)$
 $= 85900$ (2)
 b) $a = 6$, $r = \frac{1}{3}$
 $S_{\infty} = \frac{a}{1-r}$
 $= \frac{6}{1-\frac{1}{3}}$
 $= 6 \times \frac{3}{2}$
 $= 9$ (3)
 c) $y = e^{2x} + x$
 $\frac{dy}{dx} = 2e^{2x} + 1$
 when $x = 0$, $\frac{dy}{dx} = 2e^0 + 1 = 3$
 $\therefore$ slope of tangent is 3
 $y - 1 = 3(x - 0)$
 $y - 1 = 3x$
 $3x - y + 1 = 0$
 Question 6 (12)
 a) $V = \pi \int x^2 dy$
 $y = x^2 - 1$
 $x^2 = y + 1$
 $\therefore V = \pi \int_{-1}^3 (y+1) dy$
 $= \pi [\frac{1}{2}y^2 + y]_{-1}^3$
 $= \pi [\frac{1}{2}(3)^2 + 3 - \frac{1}{2}(-1)^2 - (-1)]$
 $= \pi (\frac{9}{2} + 3 - \frac{1}{2} + 1)$
 $= 8\pi \text{ units}^3$ (4)

b) (i) $W = 36$

$\therefore x + y + z = 36$

$y = 36 - 2x$ (1)

(ii) $A = xy$

$A = x(36 - 2x)$

$A = 36x - 2x^2$ (2)

(iii) $\frac{dA}{dx} = 36 - 4x$

$\frac{d^2A}{dx^2} = -4$

stationary points occur when $\frac{dA}{dx} = 0$

i.e. $36 - 4x = 0$

$4x = 36$

$x = 9$

when $x = 9$, $\frac{d^2A}{dx^2} = -4 < 0$

$\therefore$ when $x = 9$, Area is

a maximum (5)

c) (i) $x^2 + x = 2 + x - x^2$

$2x^2 - 2 = 0$

$2(x^2 - 1) = 0$

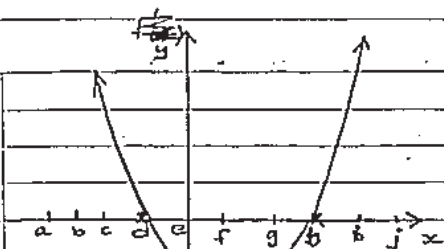
$2(x+1)(x-1) = 0$

$x = -1$ or $x = 1$

$\therefore A$ is $(-1, 0)$ and $B(1, 2)$ (3)

(ii) Area = $\int_{-1}^1 (2+x-x^2-x^2-x) dx$
 $= \int_{-1}^1 (2-2x^2) dx$
 $= [2x - \frac{2}{3}x^3]_{-1}^1$
 $= [2(1) - \frac{2}{3}(1)^3] - [2(-1) - \frac{2}{3}(-1)^3]$
 $= 2 - \frac{2}{3} + 2 - \frac{2}{3}$
 $= 2\frac{2}{3} \text{ units}^2$ (4)

(iii)



b) $\int_1^e \frac{\log x}{x} dx = \frac{1}{a} \int_a^b \frac{2 \log x}{x} dx$
 $= \frac{1}{a} [\frac{1}{2} (\log x)^2]_a^b$
 $= \frac{1}{a} (\frac{1}{2} (\log e)^2 - \frac{1}{2} (\log 1)^2)$
 $= \frac{1}{a} (\frac{1}{2} (1 - 0))$
 $= \frac{1}{2a}$ (3)

Question 6 (15)

a) $(x+3)^2 + 5(x+3) + 6 = 0$

let $m = x+3$

$m^2 + 5m + 6 = 0$

$(m+3)(m+2) = 0$

$m = -3$ or $m = -2$

$x+3 = -3$ $x+3 = -2$

$x = -6$ $x = -5$

$\therefore x = -6$ or $x = -5$ (4)

b) (i) $\alpha + \beta = \frac{-b}{a}$
 $= \frac{-\frac{1}{3}}{\frac{1}{3}}$ (1)

(ii) $\alpha\beta = \frac{c}{a}$
 $= \frac{-\frac{1}{3}}{\frac{1}{3}}$ (1)

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (\frac{1}{3})^2 - 2(-\frac{1}{3})$
 $= \frac{4}{9} + \frac{2}{3}$
 $= \frac{10}{9}$ (2)

Question 7 (15)

a) (i) $(x) f'(x) = 0$ at b, g, h (2)

$(\beta) f'(x) = 0$ at d, h (2)

$(\gamma) f''(x) = 0$ at a or f (1)

(ii) $(\alpha) f'(x) > 0$
 $a < x < d, h < x < i$ (2)

$(\beta) f''(x) < 0$
 $a < x < d$ (1)

(depends on answer above)

(b) decreasing
 $d < x < h$ (1)

Prescribed

Prescribed

Prescribed